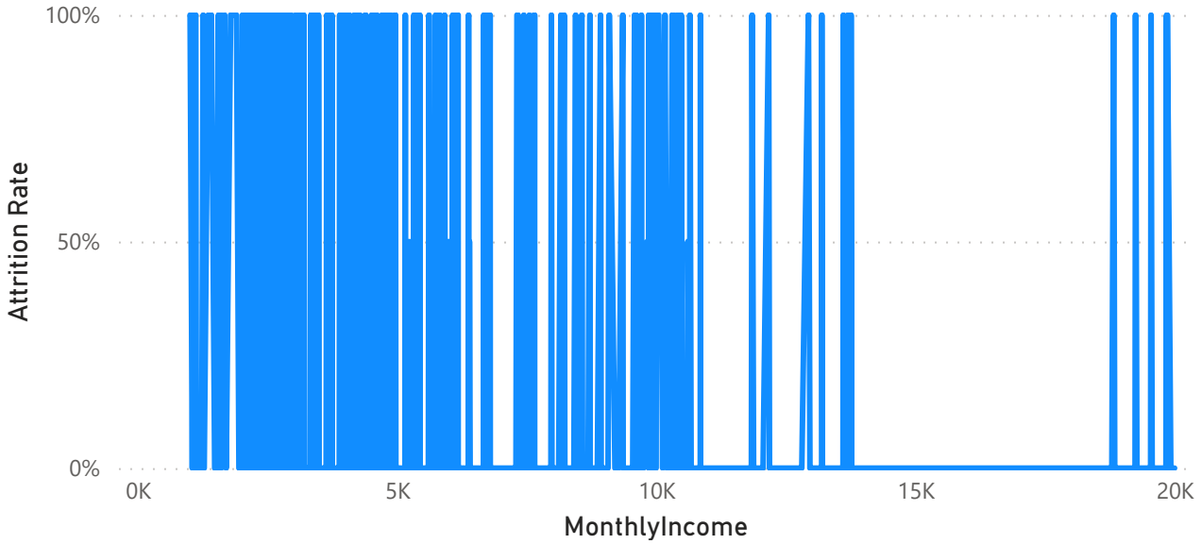
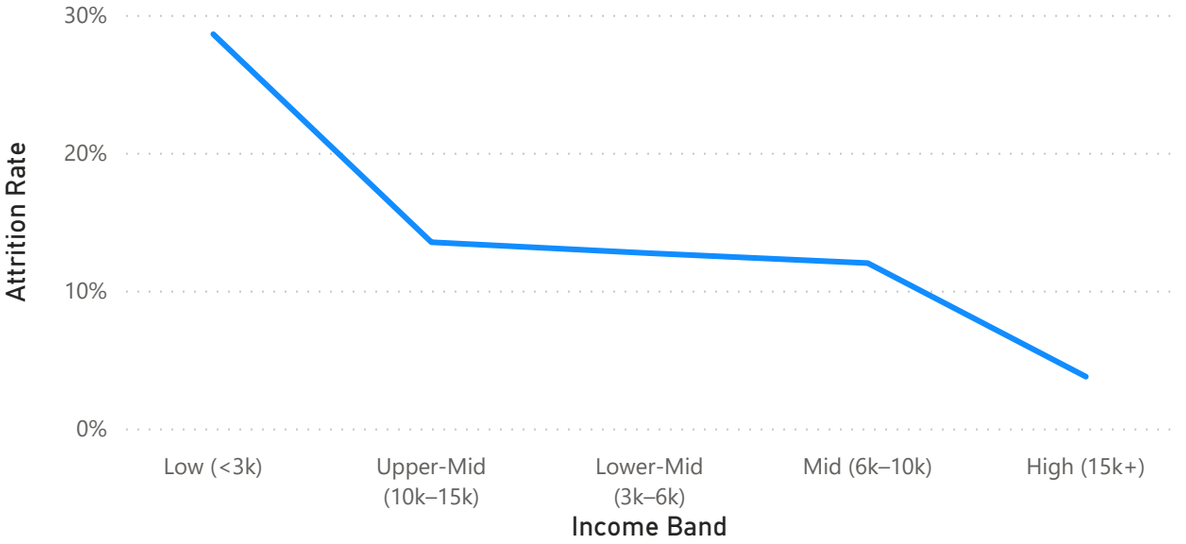
Workforce Overview & Compensation Risk Gradient

1470	237	16.12%	6,502.93
Total Employees	Employees Left	Attrition Rate	Avg Monthly Income

Attrition Rate by MonthlyIncome



Attrition Rate by Income Band (Sort by Attrition Rate Desc)

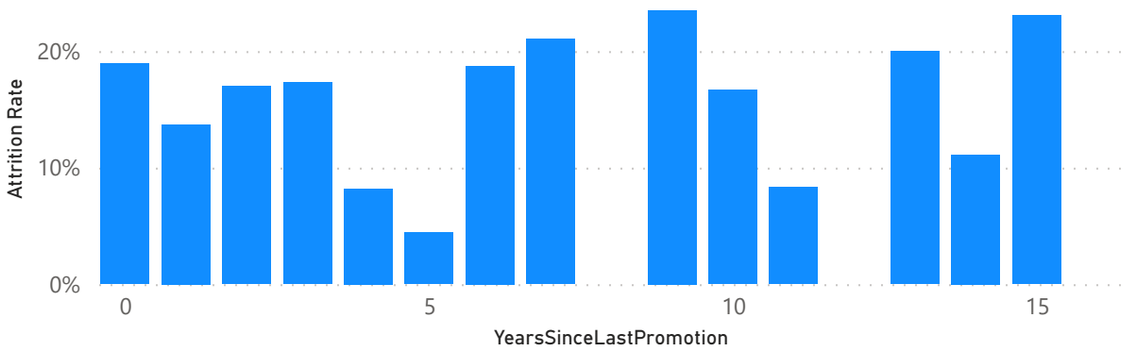


Looks noisy because each income value is its own category, so create 'Income Band' as new column in data_powerbi, so that we group the income.

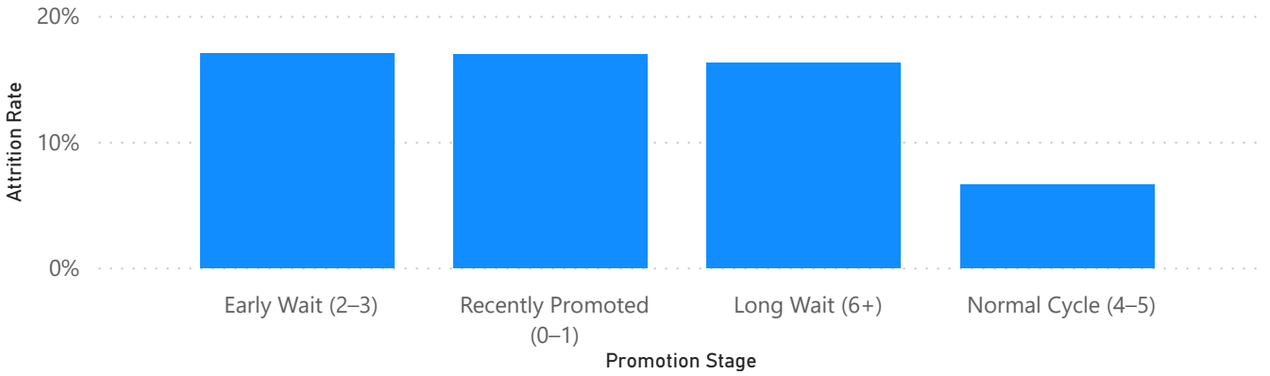
Based on the line plot, attrition risk is concentrated among lower-income employees, with rates nearly 7x higher than high-income employees. **This suggests compensation competitiveness and early-career retention strategies will deliver the greatest impact.**

Career Progression Timing & Promotion Risk Effects

Attrition Rate by YearsSinceLastPromotion

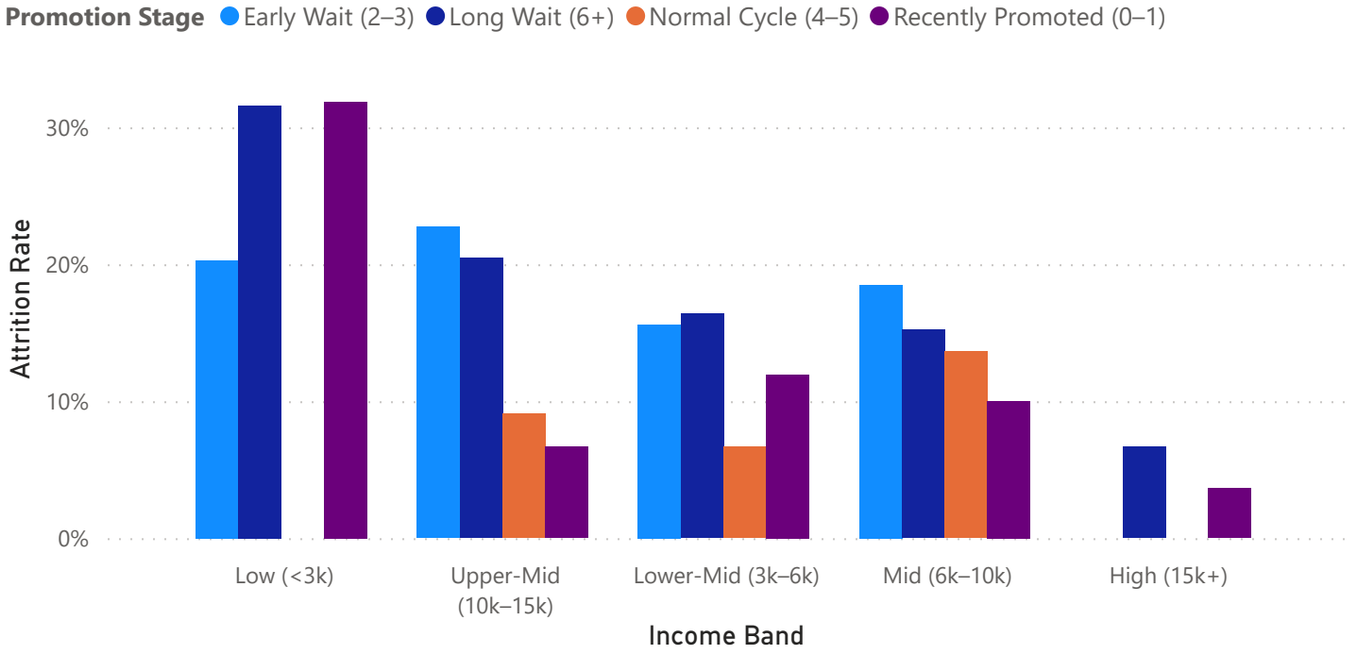


Attrition Rate by Promotion Stage



Similar to monthly income, this variable is noisy because each year is small sample, so we created promotion stage.

Attrition Rate by Income Band (Sort by Attrition Rate Desc)



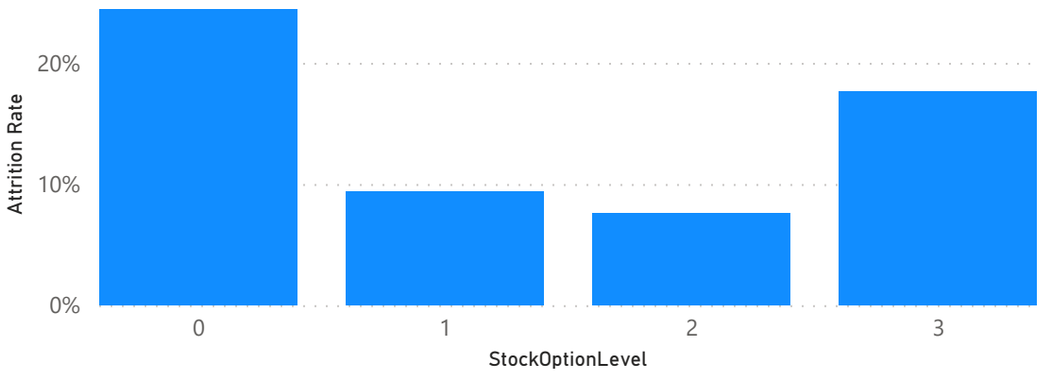
Employees in the expected promotion timing are most stable and feel career progression is fair.

Lowest attrition happens when promotion timing meets expectations ('normal cycle').

Risk increases both when employees are waiting for advancement and when they feel career progression has stalled.

Equity Incentives & Early-Career Vulnerability

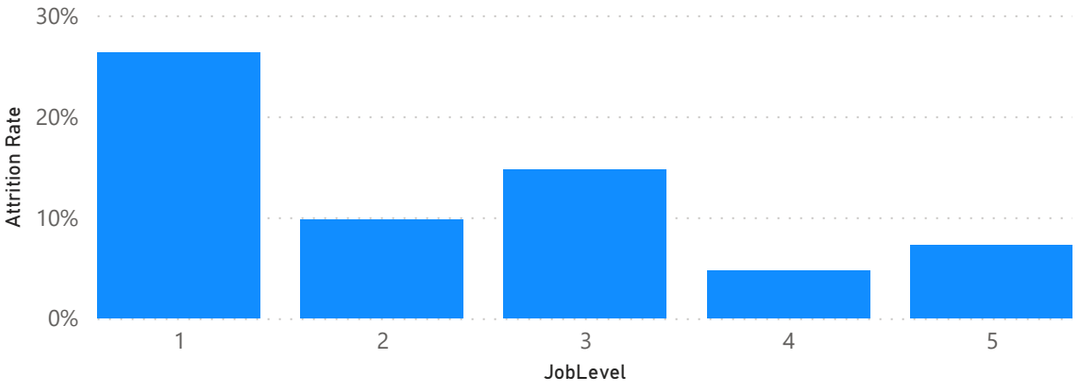
Attrition Rate by StockOptionLevel



Attrition highest at level 0
Lowest around level 2

Equity incentives correlate with retention

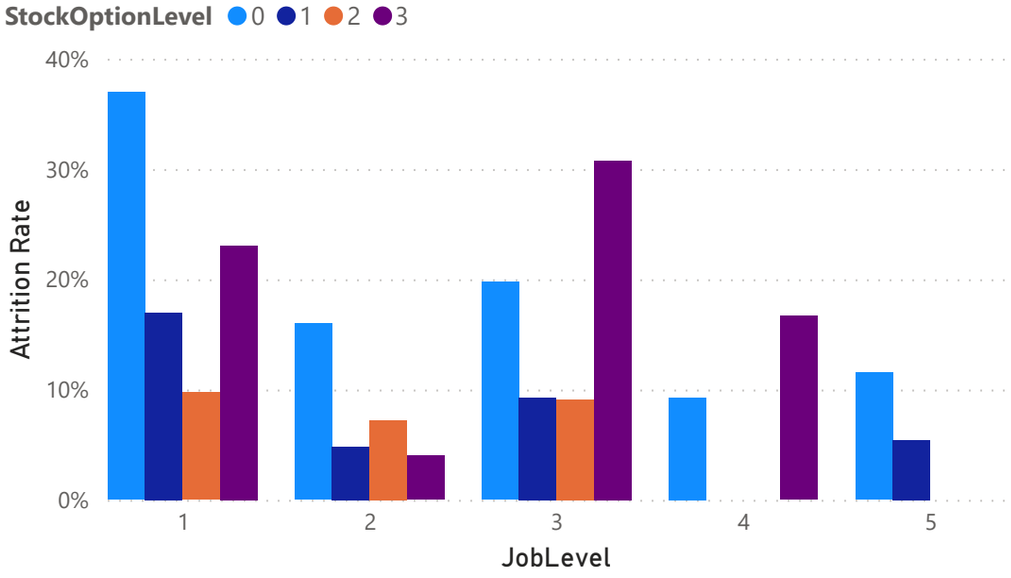
Attrition Rate by JobLevel



Attrition highest at level 1
Drops sharply at senior levels

Early career employees are most at risk

Attrition Rate by JobLevel and StockOptionLevel



Job Level 1 stands out immediately:
Employees at Job Level 1 with Stock Option Level 0 have the highest attrition (~0.38)
This is a very strong signal: **Early-career employees without equity incentives are the most at risk of leaving.**

Stock options appear to reduce attrition (especially early career)
Within Job Level 1:
Level 0 → very high attrition
Level 1 / 2 → noticeably lower
Level 3 → much lower
This suggests a protective effect of equity

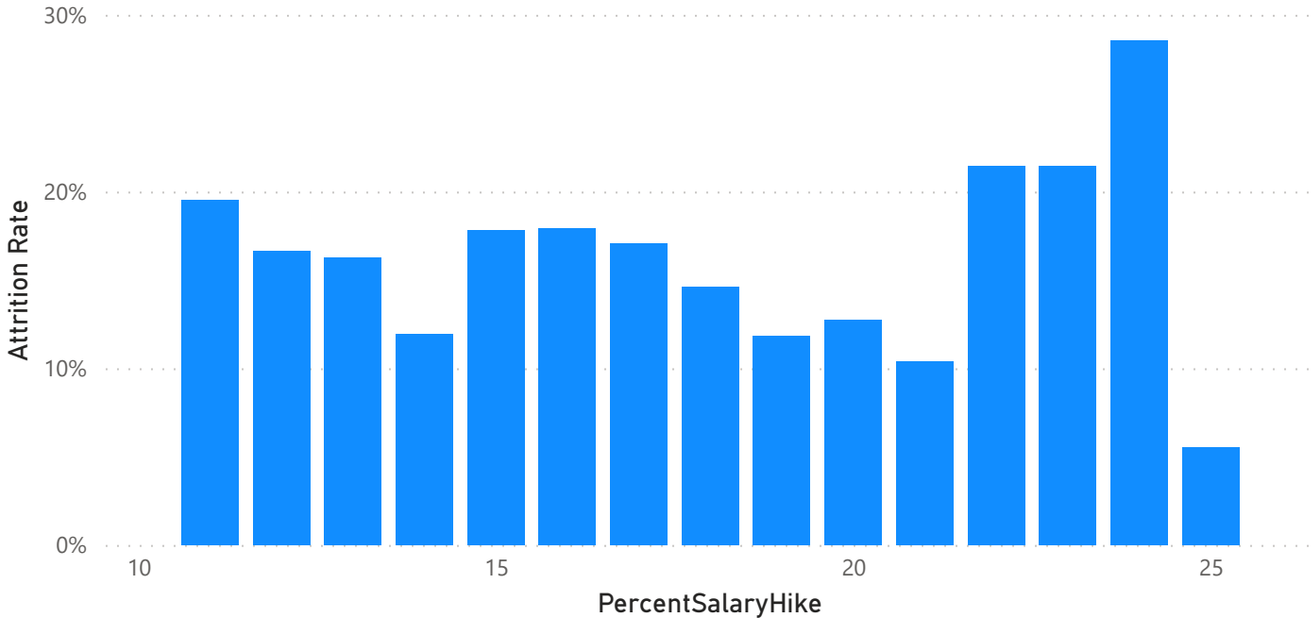
Mid levels show mixed patterns: Job Levels 2 and 3 still show variation, but differences are smaller

Senior levels (4–5) have low attrition overall, even with low stock options, attrition stays low

Attrition risk is concentrated among junior employees without stock incentives, suggesting compensation structure plays a critical role in early-career retention.

Salary Growth Alone Is Insufficient to Drive Retention

Attrition Rate by PercentSalaryHike



Attrition shows a non-linear relationship with salary increases.
Employees receiving very small or very large raises tend to have higher attrition, while moderate increases are associated with slightly lower turnover.

This suggests that pay adjustments alone do not drive retention.
Instead, salary growth likely works in combination with career progression and role level.

Workload & Travel Pressure as Behavioral Risk Drivers

Age Group

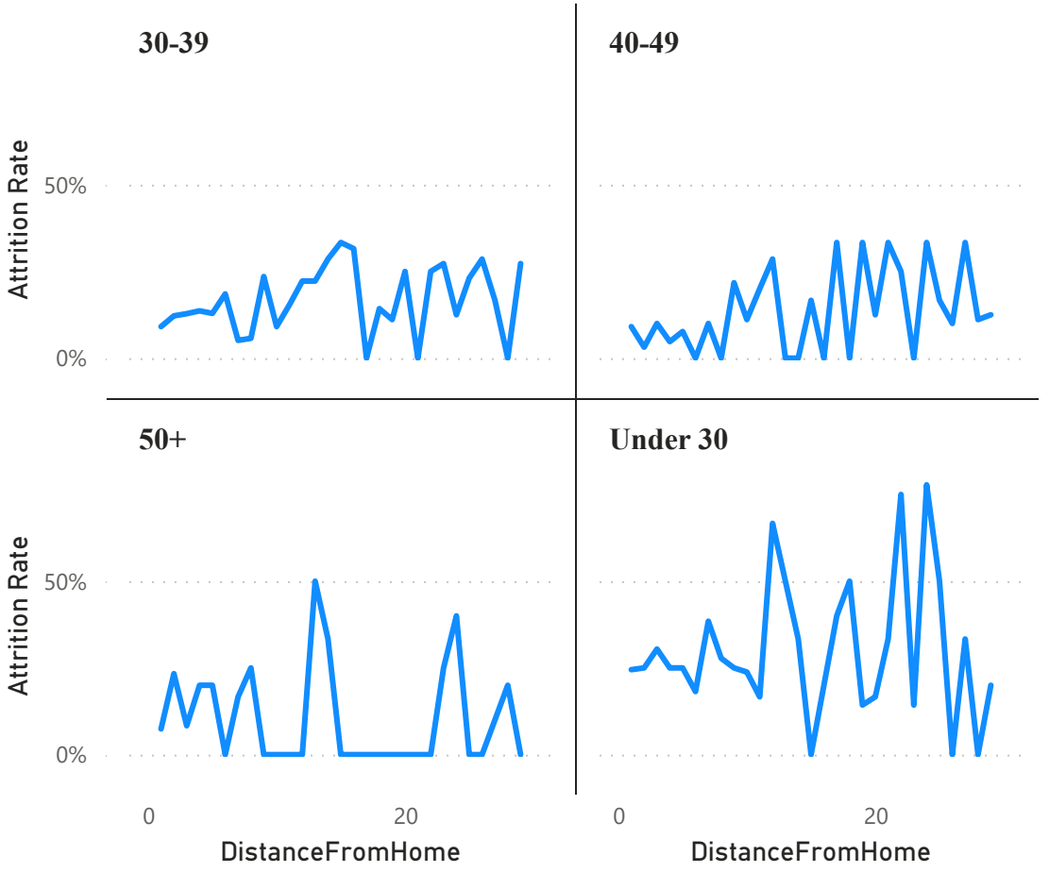
All

30.53%
Attrition Rate OT Yes

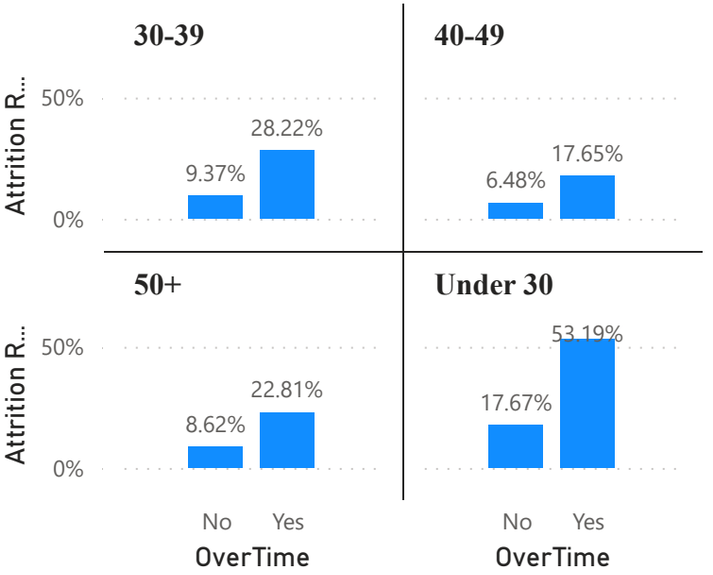
10.44%
Attrition Rate OT No

20.09%
Marginal Increase OT

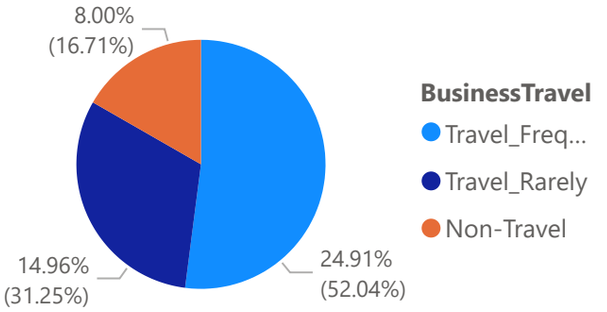
Attrition Rate by DistanceFromHome and Age Group



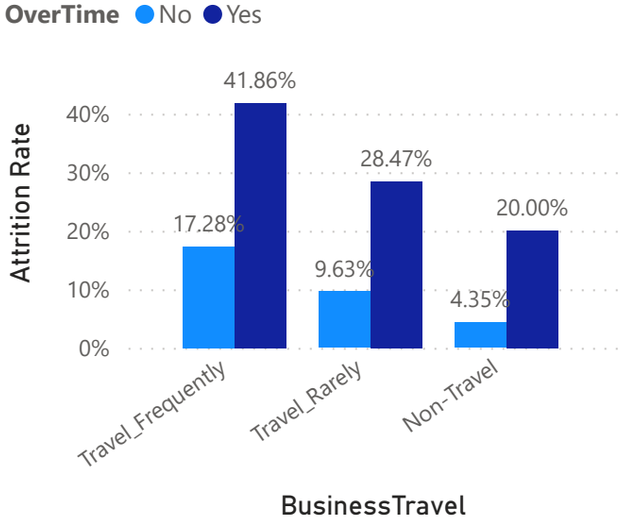
Attrition Rate by OverTime and Age Group



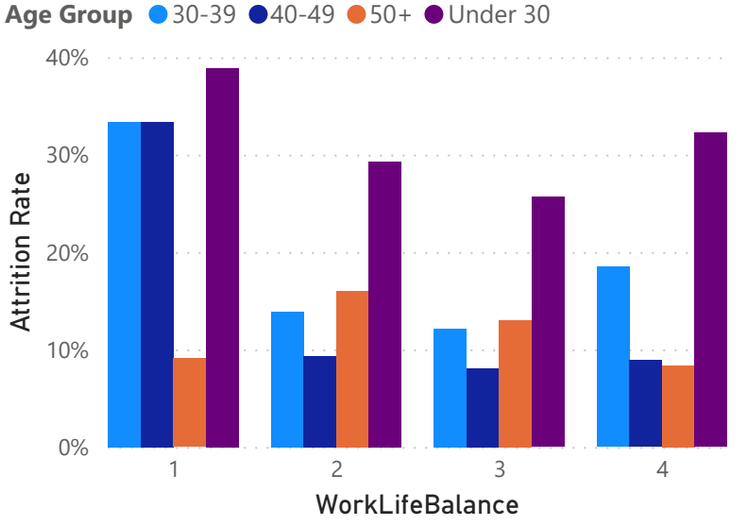
Attrition Rate by BusinessTravel



Attrition Rate by BusinessTravel and OverTime



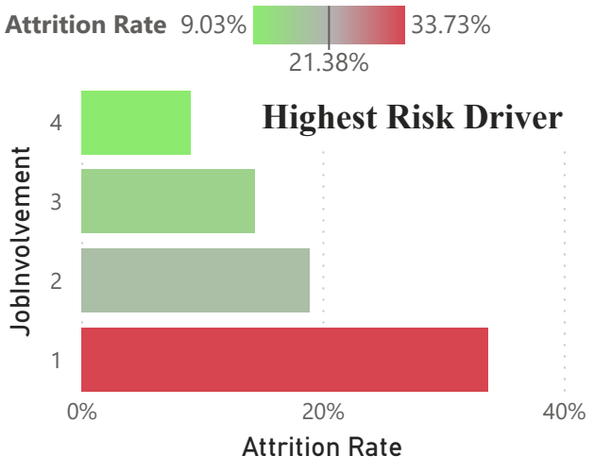
Attrition Rate by WorkLifeBalance and Age Group



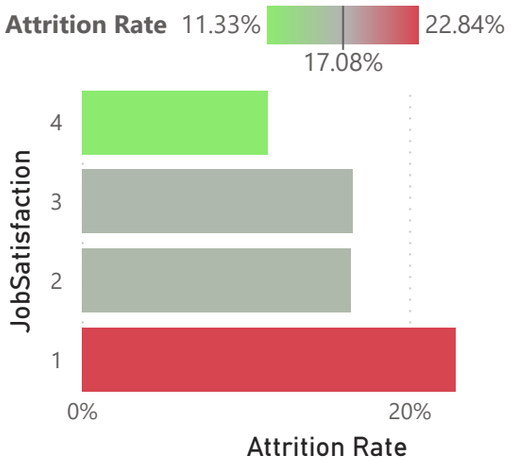
Engagement Deficit as the Primary Attrition Driver

Overall Attrition Rate	Low Involvement Attrition	JobInvolvement Risk Gap	Low Job Involvement Workforce
16.12%	33.73%	24.71%	31.16%

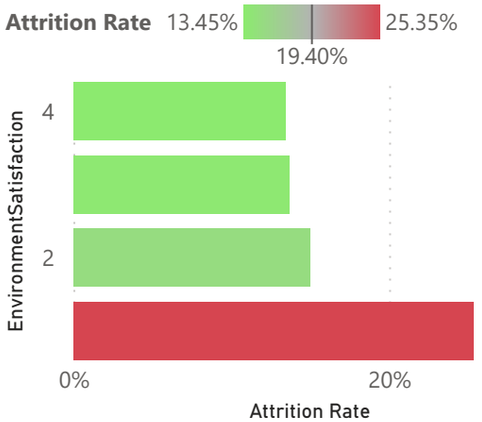
Attrition Rate by JobInvolvement



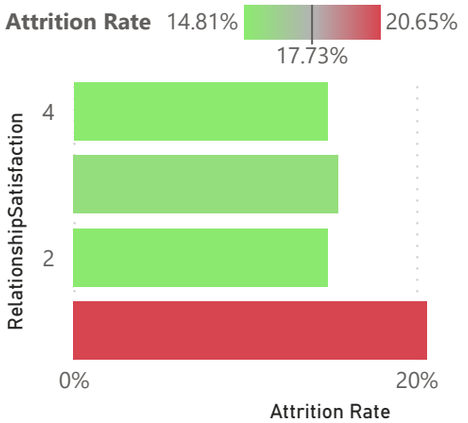
Attrition Rate by JobSatisfaction



Attrition Rate by EnvironmentSatisfaction



Attrition Rate by RelationshipSatisfaction



SatisfactionType	1	2	3	4
Environment Satisfaction	25.35%	14.98%	13.69%	13.45%
Job Involvement	33.73%	18.93%	14.40%	9.03%
Job Satisfaction	22.84%	16.43%	16.52%	11.33%
Relationship Satisfaction	20.65%	14.85%	15.47%	14.81%

Key Insights:

- Job involvement exhibits the steepest attrition gradient (24.7pp differential).
- 31% of employees fall into low job involvement levels, representing concentrated risk.
- Targeting Level 1–2 employees offers the highest-impact retention opportunity.

Department	JobRole	Attrition
All	All	All

Attrition Concentration by Department, Role & Training Investment

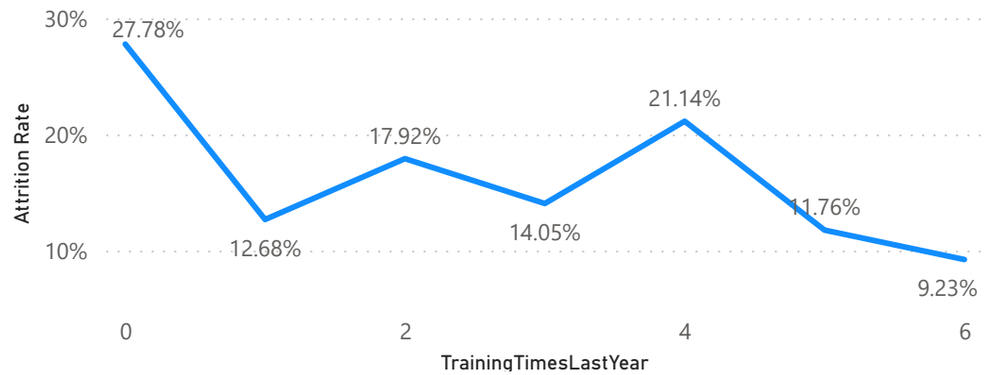
1470 Total Employees

16.12% Attrition Rate

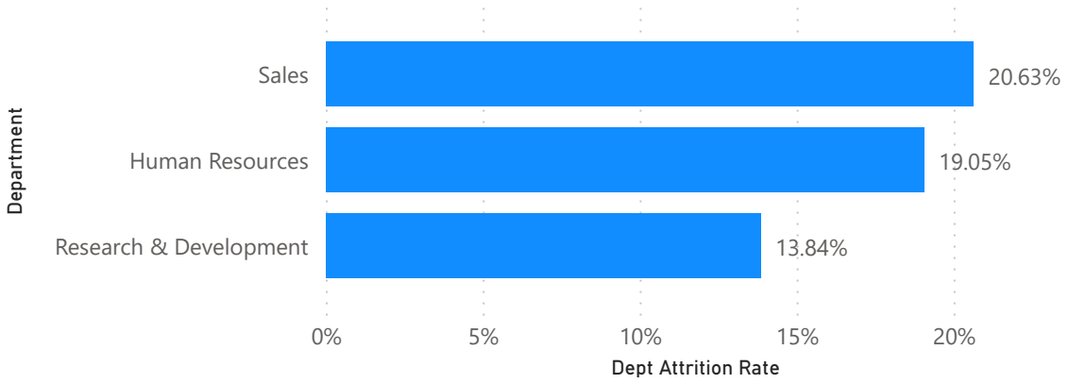
237 Employees Left

6.50K Avg Monthly Income

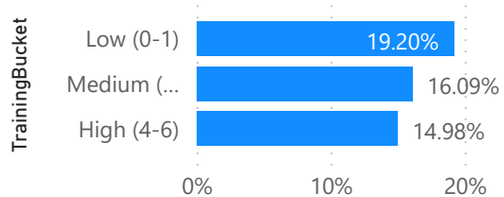
Attrition Rate by TrainingTimesLastYear



Dept Attrition Rate by Department



Attrition Rate by TrainingBucket



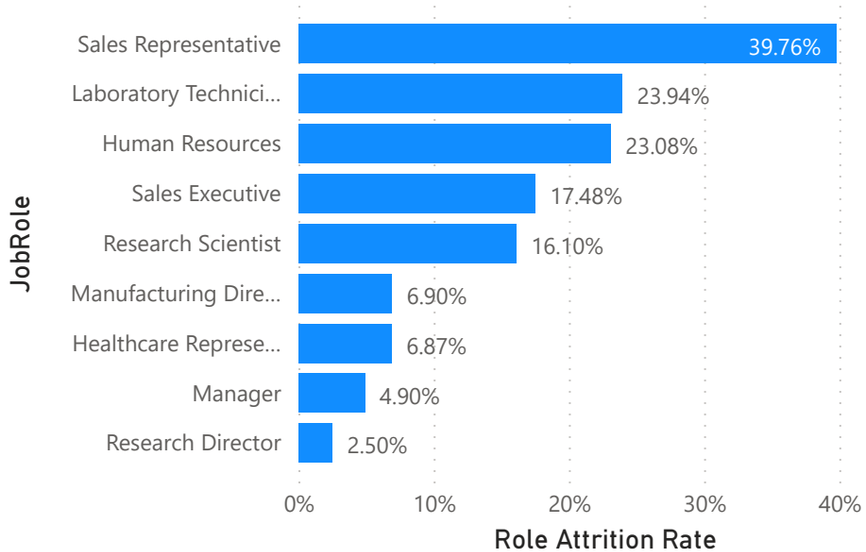
TrainingBucket

- ☐ High (4-6)
- ☐ Low (0-1)
- ☐ Medium (2-3)

Department

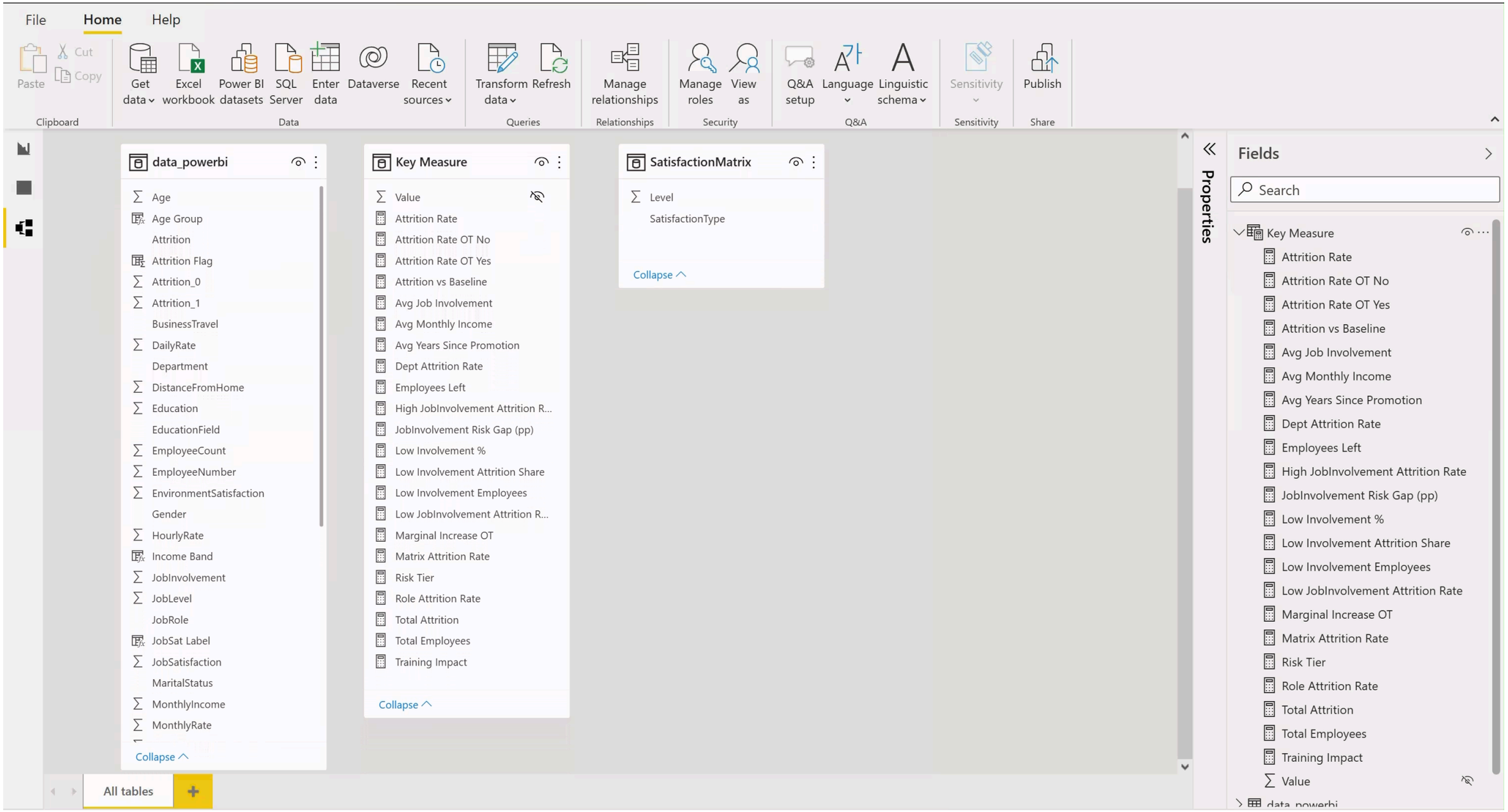
- ☐ Human Resources
- ☐ Research & Development
- ☐ Sales

Role Attrition Rate by JobRole



JobRole	High (4-6)	Low (0-1)	Medium (2-3)	Total
Sales Representative	33.33%	25.00%	42.62%	**39.76%**
Sales Executive	16.18%	22.22%	17.32%	**17.48%**
Research Scientist	21.43%	13.33%	15.05%	**16.10%**
Research Director	0.00%	12.50%	1.79%	**2.50%**
Manufacturing Director	9.68%	11.76%	5.15%	**6.90%**
Manager	0.00%	0.00%	6.33%	**4.90%**
Laboratory Technician	19.05%	42.86%	24.18%	**23.94%**
Human Resources	12.50%	42.86%	21.62%	**23.08%**
Healthcare Representative	3.57%	7.14%	7.87%	**6.87%**
Total	**14.98%**	**19.20%**	**16.09%**	**16.12%**

JobRole

- ☐ Healthcare Representative
- ☐ Human Resources
- ☐ Laboratory Technician
- ☐ Manager
- ☐ Manufacturing Director
- ☐ Research Director
- ☐ Research Scientist
- ☐ Sales Executive
- ☐ Sales Representative

Data Model Architecture & Measure Framework